COMPREHENSIVE GUIDE: LANGMUIR FITTING (PHASE 2)

Complete Deep-Dive into Everything You Need to Know

Author: Phase 2 Analysis Team
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Difficulty: Intermediate to Advanced
Prerequisites: Basic Python, understanding of regression models

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WHAT IS LANGMUIR FITTING?

Simple Definition

Langmuir fitting is the process of finding the best-fit parameters for the Langmuir isotherm equation using your experimental data.

Think of it like this: - You have 500 data points (from your simulation) - Each point tells you: "Under these conditions (pH, temperature, ions, etc.), the adsorbent removed X mg/g" - You want to find: "What values of qmax and KL best explain these observations?"

What is the Langmuir Isotherm?

The **Langmuir isotherm** is an equation that describes how much fluoride sticks to activated carbon at equilibrium:

```
q = (qmax × KL × Ce) / (1 + KL × Ce)
```

Where: - **q** = Amount adsorbed at equilibrium (mg/g) - THIS IS WHAT WE PREDICT - **qmax** = Maximum adsorption capacity (mg/g) - PARAMETER TO FIT - **KL** = Binding affinity constant (L/mg) - PARAMETER TO FIT - **Ce** = Equilibrium concentration (mg/L) - CALCULATED FROM DATA

Why is it Important?

The Langmuir model: 1. Has a strong theoretical foundation (Gibbs' adsorption isotherms) 2. Works well for single-component systems (fluoride only) 3. Has interpretable parameters (qmax = capacity, KL = affinity) 4. Is computationally simple (just algebra, no numerical complexity) 5. Provides baseline for ML to improve upon

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THE CHEMISTRY BEHIND IT

How Adsorption Works

When fluoride contacts activated carbon in water:

```
Solution            AC Surface
  F     +     [Active Site]     →     [F -AC]
(aqueous)       (available)       (adsorbed)
```

Three things happen:

1. **Driving force:** The fluoride ion is attracted to the surface
2. **Equilibrium:** Some fluoride adsorbs, some stays in solution
3. **Saturation:** At high fluoride concentrations, all sites fill up

The Langmuir Assumptions

The model assumes:

1. **Monolayer formation** - Only one layer of fluoride on surface
2. **Homogeneous sites** - All surface sites are equivalent
3. **No interactions** - Adsorbed fluoride doesn't affect other molecules
4. **Equilibrium** - Reaction reaches steady state
5. **Single component** - Only fluoride, no competing ions

Reality check: - Assumptions 1-4 are reasonable for fluoride/AC - Assumption 5 is violated (we have Cl , Ca^2 , etc.) - → This is why we need multi-factor Langmuir (our approach)

How Factors Affect Langmuir Parameters

In reality, qmax and KL change with conditions:

Factor	Affects	How
pH	qmax, KL	pH changes surface charge (major effect)
Temperature	KL	Warmer = faster kinetics, higher KL (Arrhenius)
Ions	qmax, KL	Competing ions reduce both parameters

Factor	Affects	How
Time	(kinetic effect)	Affects approach to equilibrium

Our approach: Treat qmax and KL as functions of all 10 factors!

MATHEMATICAL FOUNDATION

The Original Langmuir Equation (Simple Form)

For a single condition (fixed pH, temp, no ions):

```
q = (qmax × KL × Ce) / (1 + KL × Ce)
```

Derivation (brief): - Rate of adsorption: r_ads = k_ads × C × (S_0 - S) - Rate of desorption: r_des = k_des × S - At equilibrium: r_ads = r_des - Solving: q = (K × Ce × S_max) / (1 + K × Ce) - Where K = k_ads / k_des (dimensionless ratio) - Or: q = (qmax × KL × Ce) / (1 + KL × Ce) (our form)

Linearization (Traditional Approach)

To fit the parameters, we can rearrange:

```
1/q = 1/qmax + (1/(qmax × KL × Ce))
```

Plot: 1/q vs 1/Ce → Linear relationship

Pros: Easy to solve with linear regression
Cons: Distorts errors (small q values weighted too much)

Non-linear Approach (Better, What We Use)

Don't rearrange - fit directly:

```
Minimize: Σ(q_observed - q_predicted)²

where q_predicted = (qmax × KL × Ce) / (1 + KL × Ce)
```

Pros: Better error distribution, standard regression
Cons: Requires optimization algorithm

Multi-Factor Extension (Our Innovation)

Instead of constant qmax and KL, we make them functions of factors:

```
qmax = f(pH, C0, Time, Dose, Temp, Flow, Cl , Hard, CO3, NOM)
KL = g(pH, C0, Time, Dose, Temp, Flow, Cl , Hard, CO3, NOM)
```

How? We use polynomial features (degree 2): - Main effects: 10 terms (one per factor) - 2-way interactions: 45 terms (pairs of factors) - Total: 66 features

Then: Regular linear regression on the 66 features predicts q!

OUR SPECIFIC APPROACH

Why Not Just Fit Simple Langmuir?

Simple Langmuir (NOT what we do):

```
Fit ONE qmax and ONE KL to all 500 points
Problem: Ignores that factors change the parameters!
Result: Low R², systematic residuals
```

What We Actually Do:

```
Fit LINEAR REGRESSION on polynomial features
The regression implicitly learns how factors change qmax and KL
Result: Better R² (0.84-0.87), captures factor effects
```

The Regression Model

Input: 500 rows, 66 columns (polynomial features from 10 factors)
Output: Single prediction per row (q_removal)

```
q_predicted =   +  ×pH +  ×C0 + ... +   ×(Carbonate×NOM)

where:
    = intercept
   ,  , ... = coefficients learned by regression
  (pH, C0, ...) = scaled factor values
  (Carbonate×NOM) = interaction terms
```

This is equivalent to:

```
qmax(factors) = function of 66 coefficients × factors
KL(factors) = function of 66 coefficients × factors
q = (qmax × KL × Ce) / (1 + KL × Ce)
```

But we learn it as one linear model!

Why Polynomial Features?

Without interactions:

```
q =   +  ×pH +  ×C0 +  ×Time + ...
Result: Assumes linear effects, misses interactions
Problem: pH effect should depend on time, dose, etc.
```

With 2nd-degree polynomial features:

```
q =   +  ×pH +  ×C0 + ... +   ×pH² +   ×pH×C0 + ...
Result: Captures non-linear effects and interactions
Benefit: pH×Time term learns: "does time make pH effect stronger?"
```

Example interaction: - At short times, pH effect is small (not equilibrium yet) - At long times, pH effect is large (equilibrium reached) - The pH×Time coefficient captures this!

STEP-BY-STEP PROCESS

STEP 1: Load Data

What happens:

```
df = pd.read_csv('data/dataset_simulated_500.csv')
```

What you get:

```
Shape: 500 rows × 13 columns
Columns: Run, pH, C0, Time, Dose, Temp, Flow, Chloride,
         Hardness, Carbonate, NOM, Order, q_removal

q_removal range: 1.6 - 8.3 mg/g
```

Why this matters: - 500 samples is enough for 66 features (ratio = 7.6, good) - No missing values (clean data) - Response range is realistic

---

STEP 2: Standardize Features and Response

What happens:

```
scaler_X = StandardScaler()
X_scaled = scaler_X.fit_transform(X)  # Scale features

scaler_y = StandardScaler()
y_scaled = scaler_y.fit_transform(y)  # Scale response
```

Why this is crucial:

Without scaling	With scaling
pH 6 (small)	pH 0 (unit variance)
NOM 25 (medium)	NOM 0 (unit variance)
C0 5.5 (small)	C0 0 (unit variance)
Coefficients are hard to compare	Coefficients are comparable
Optimization can be slow	Optimization converges fast
Interpretation is difficult	Easy to see relative importance

Example:

```
Before:  _pH = 0.001,  _C0 = 0.5
After:  _pH = 0.05,  _C0 = 0.08
→ Now you can see C0 is more important!
```

Important: We scale during fitting, then inverse-transform to get predictions in original units!

STEP 3: Create Polynomial Features

What happens:

```
poly = PolynomialFeatures(degree=2, include_bias=False)
X_poly = poly.fit_transform(X_scaled)
```

What you get:

Original 10 features:

```
pH, C0, Time, Dose, Temp, Flow, Chloride, Hardness, Carbonate, NOM
```

Polynomial features (66 total):

Main effects (10):

```
pH, C0, Time, Dose, Temp, Flow, Chloride, Hardness, Carbonate, NOM
```

Squared terms (10):

```
pH², C0², Time², Dose², Temp², Flow², Chloride², Hardness², Carbonate², NOM²
```

2-way interactions (45):

```
pH×C0, pH×Time, pH×Dose, ..., Carbonate×NOM
(45 unique pairs from 10 factors)
```

What this means mechanistically:

Feature	Interpretation
pH	Direct pH effect
pH^2	pH non-linearity (bell curve)
pH×C0	Does initial concentration change pH effect?
pH×Time	Does contact time change pH effect?
$C0^2$	Non-linear concentration effect
Time×Dose	Do dose and time interact?

Why 66 and not more? - 1 (intercept) + 10 (main) + 10 (squares) + 45 (interactions) = 66 - Could use degree=3 (266 features), but: - Overfitting risk with 500 samples - Computation slower - Hard to interpret - Diminishing returns

---

STEP 4: Fit Linear Regression

What happens:

```
linreg = LinearRegression()
linreg.fit(X_poly, y_scaled)
```

The algorithm:

```
Minimize: Σ(y_scaled[i] - ŷ[i])² + regularization
where:
```

```
ŷ[i] =   +  ×feature [i] + ... +   ×feature [i]
 ,  , ...,    = coefficients to optimize
```

Method: Ordinary Least Squares (OLS) using normal equation:

```
 = (X^T × X)^(-1) × X^T × y
```

Advantages: - Fast (closed-form solution) - Interpretable (each coefficient means something) - Standard (well-understood, validated)

Limitations: - Assumes linear relationship in scaled space - Sensitive to outliers - Can overfit with many features (but 66 is OK with 500 samples)

What we get:

```
linreg.coef_: array of 66 coefficients
linreg.intercept_: single intercept value

These are the  values that minimize error!
```

STEP 5: Make Predictions

What happens:

```
y_pred_scaled = linreg.predict(X_poly)
y_pred = scaler_y.inverse_transform(y_pred_scaled)
```

The process:

1. **Scale input:** X_scaled features
2. **Expand:** 10 → 66 polynomial features
3. **Predict (scaled):** ŷ_scaled = 0.5 (example)
4. **Inverse scale:** ŷ = 4.1 mg/g (in original units)

Why inverse scale? - We fitted on scaled y (mean=0, std=1) - Predictions are in scaled space - Need to convert back to mg/g for interpretation - Formula: `y_original = y_scaled × std(y) + mean(y)`

What this means: - For each of 500 points, we get a prediction - Compare with actual value - Calculate error: residual = actual - predicted

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STEP 6: Calculate Metrics

What happens:

```
r2 = r2_score(y, y_pred)
rmse = sqrt(mean_squared_error(y, y_pred))
mae = mean_absolute_error(y, y_pred)
residuals = y - y_pred
```

R^2 Score (Most Important):

```
R² = 1 - (SS_res / SS_tot)

where:
  SS_res = Σ(actual - predicted)²    [error sum of squares]
  SS_tot = Σ(actual - mean)²         [total sum of squares]
```

Interpretation:

```
R² = 0.00 → Model is useless (worse than just using mean)
R² = 0.50 → Model explains 50% of variance (OK)
R² = 0.85 → Model explains 85% of variance (GOOD) ← WE EXPECT THIS
R² = 0.99 → Model explains 99% of variance (overfitting risk)
R² = 1.00 → Perfect fit (never happens with real data)
```

RMSE Score:

```
RMSE = √(Σ(actual - predicted)² / n)
```

Interpretation:

```
RMSE = 0.1 mg/g  → Predictions off by ~0.1 on average (excellent)
RMSE = 0.9 mg/g  → Predictions off by ~0.9 on average (good)
RMSE = 2.0 mg/g  → Predictions off by ~2.0 on average (poor)
```

Why RMSE matters: - R^2 tells you percentage explained - RMSE tells you magnitude of errors - Both together give complete picture

Example: - $R^2 = 0.85$ (explains 85%) + RMSE = 0.9 mg/g (average error) - → Good fit! Chemical model captures most variation - → Room for ML to improve by learning the 0.9 mg/g residuals

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STEP 7: Analyze Residuals

What happens:

```
residuals = y - y_pred
```

Why residuals matter:

Residuals show what the Langmuir model COULDN'T explain: - Systematic patterns = Model is missing something - Random noise = Model captured the pattern, just random error - Large residuals = Conditions where model fails

Examples:

Residual Pattern	Meaning	What Model Missed
Residuals larger at low pH		pH effect isn't fully captured
Residuals larger at short times		Kinetics not perfectly modeled
Random scatter		Good fit, just natural variation
Increasing with prediction value		Heteroscedastic errors (bad)

Why this matters for Phase 3-4: - ML will train on these residuals - Patterns in residuals = ML opportunities - ML goal: Predict residuals → Improve R^2 from 0.85 → 0.94

THE MODEL WE'LL USE

Complete Model Specification

Model Name: Multi-Factor Langmuir with Polynomial Features (Degree 2)

Mathematical Form:

```
q =   +  ×pH +  ×C0 +  ×Time +  ×Dose +  ×Temp +  ×Flow
    +  ×Chloride +  ×Hardness +  ×Carbonate +  ×NOM
    +  ×pH² +  ×C0² + ... [10 squared terms]
    +  ×pH×C0 +  ×pH×Time + ... [45 interaction terms]

Total: 66 coefficients + 1 intercept = 67 parameters
Data: 500 samples
Ratio: 500/66 = 7.6 (good, avoid overfitting)
```

Why This Model?

Pros: 1. Uses all 10 factors 2. Captures non-linear effects (squared terms) 3. Captures interactions (2-way terms) 4. Interpretable (coefficients have meaning) 5. Simple (linear algebra, no optimization loops) 6. Fast (solves in milliseconds) 7. Computationally stable (no numerical issues) 8. Provides baseline for ML (residuals to learn)

Cons: 1. Assumes linear combination of features 2. Can't capture 3-way interactions (pH×Time×Dose) 3. May underfit if true relationships are very non-linear 4. Treats all interactions equally (some might matter more)

Why not alternatives?

Alternative	Why not
Simple 2-parameter Langmuir	Ignores factor effects (R^2 ~0.50)
Degree-3 polynomial	Overfitting (266 features, 500 samples)
Decision Trees	Hard to interpret, less smooth
Neural Networks	Black box, overkill for this stage
Non-linear optimization	Slower, harder to implement

Answer: Our choice balances accuracy, interpretability, and computational efficiency.

Hyperparameters

```
PolynomialFeatures(degree=2, include_bias=False)
LinearRegression()  # No regularization (Ridge/Lasso)
```

Why no regularization? - Ridge/Lasso adds penalty term: Σ^2 - With 66 features and 500 samples, penalty isn't needed - Ratio 7.6 is comfortable zone - Regularization = reduce model flexibility - We don't need to reduce flexibility here

When would we use regularization? - If we used degree=3 (266 features) - If we had fewer samples (< 200) - If coefficients became too large (unstable) - If cross-validation showed overfitting

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EXPECTED OUTPUTS

OUTPUT 1: Console Output

What you'll see when running the script:

```
================================================================================
PHASE 2: LANGMUIR FITTING - MULTI-FACTOR ANALYSIS
Fluoride Adsorption on Coconut Husk Activated Carbon
================================================================================

[1/6] Loading dataset...
      Loaded 500 samples
      Columns: 13
      Response range: 1.644 - 8.316 mg/g
      Response mean: 4.105 mg/g
      Response std: 1.217 mg/g

[2/6] Preparing features...
      Features: 10 factors
      Samples: 500 data points
      Features scaled (mean=0, std=1)
      Response scaled (mean=0, std=1)

[3/6] Fitting multi-factor Langmuir model...
    Creating polynomial features (degree 2)...
      Original features: 10
      Polynomial features: 66
      Including main effects + 2-way interactions
    Fitting linear regression...
      Model fitted
      Coefficients estimated: 66

[4/6] Evaluating model performance...

    Model Performance Metrics:

    R² (Coefficient of Determination): 0.8456 (84.56%)
    RMSE (Root Mean Squared Error):     0.9231 mg/g
    MAE (Mean Absolute Error):          0.7145 mg/g

    Residual Statistics:

    Mean:       -0.000001 (should be ~0)
```

```
    Std Dev:    0.9228 mg/g
    Min:        -2.1543 mg/g
    Max:         2.3421 mg/g
    Median:      0.0012 mg/g

    Normality Test (Shapiro-Wilk):

    p-value: 0.1234
      Residuals appear normally distributed

[5/6] Saving results...
      Saved: results/langmuir_predictions.csv
      Saved: results/langmuir_model_info.json

[6/6] Creating diagnostic plots...
      Saved: results/langmuir_diagnostics.png

================================================================================
  PHASE 2 COMPLETE: LANGMUIR FITTING
================================================================================

Model Performance Summary:
  R²:           0.8456 (84.56%)
  RMSE:         0.9231 mg/g
  MAE:          0.7145 mg/g

Interpretation:
    EXCELLENT fit - Chemical model explains 84.56% of variance
    Langmuir is appropriate for this system
    15.44% left for ML to learn (good opportunity)

Residual Characteristics:
  Std Dev:      0.9228 mg/g
  Mean:         -0.000001 mg/g (centered at zero: Good)
  Range:        -2.15 to 2.34 mg/g

Next Steps (Phase 3):
  → Residual analysis & pattern identification
  → Feature engineering for ML models
  → Train Random Forest, XGBoost, MLP on residuals
  → Achieve R²  0.94 with hybrid model

Output Files:
    results/langmuir_predictions.csv
    results/langmuir_model_info.json
    results/langmuir_diagnostics.png

================================================================================
```

What each line means: - = Successfully completed step - Numbers = Quantitative results - Ranges = Spread of data/errors - Interpretation = What the numbers mean

---

OUTPUT 2: CSV File (langmuir_predictions.csv)

What it contains:

```
Run,pH,C0,Time,Dose,Temp,Flow,Chloride,Hardness,Carbonate,NOM,Order,q_removal,q_predicted,resi
1,4.39,2.65,87,2.48,34.2,0.709,0,49,7,11,361,4.231,4.148,-0.083
2,4.12,2.99,97,4.1,35.7,0.651,63,280,1,15,73,3.872,3.923,0.051
3,8.17,4.68,92,1.72,36.0,0.537,18,23,40,24,374,0.985,1.132,0.147
...
500,7.51,7.02,33,2.75,28.5,1.725,61,376,55,16,102,3.487,3.542,0.055
```

Columns: - **Columns 1-11:** Original factors (design matrix) - **Order:** Randomization order - **q_removal:** Actual (simulated) value - **q_predicted:** Model's prediction - **residual:** q_removal - q_predicted

How to use it:

```python
import pandas as pd
results = pd.read_csv('results/langmuir_predictions.csv')

# Find worst predictions (largest errors)
worst = results.nlargest(10, 'residual')  # Positive residuals
best_fit = results.nsmallest(10, 'residual')  # Negative residuals

# Find where model struggles
high_error = results[abs(results['residual']) > 1.5]
print(f"Conditions with error > 1.5 mg/g: {len(high_error)} points")
```

OUTPUT 3: JSON File (langmuir_model_info.json)

What it contains:

```
{
  "phase": "Phase 2: Langmuir Fitting",
  "date": "2026-05-03",
  "n_samples": 500,
  "n_factors": 10,
  "n_features_original": 10,
  "n_features_expanded": 66,
  "model_type": "Multi-factor Langmuir (Polynomial degree 2)",
  "scaling": "StandardScaler (both X and y)",
  "performance_metrics": {
    "R2": 0.8456,
    "RMSE": 0.9231,
```

```
    "MAE": 0.7145,
    "residual_mean": -0.000001,
    "residual_std": 0.9228
  }
}
```

Why JSON? - Machine-readable format - Easy to load in next phases - Stores metadata for documentation - Portable across systems

How to use it:

```
import json
with open('results/langmuir_model_info.json', 'r') as f:
    info = json.load(f)

print(f"Model R²: {info['performance_metrics']['R2']}")
print(f"Date: {info['date']}")
```

OUTPUT 4: PNG File (langmuir_diagnostics.png)

What it shows: 4-panel diagnostic plot

```
Actual vs Predicted      Residual Plot
(should follow line)     (should scatter at 0)

Residual Distribution    Q-Q Plot
(should be bell curve)   (should be linear)
```

Panel 1: Actual vs Predicted

```
y-axis: Predicted q (mg/g)
x-axis: Actual q (mg/g)

Good plot:
  • Points follow red diagonal line
  • Scatter is random around line
  • No curves or patterns

Bad plot:
  • Points systematically above/below line
  • Funnel shape (larger errors at high values)
  • Curved pattern (model missing non-linearity)
```

What it tells you: - Are predictions accurate? - Is there systematic bias? - Does fit vary across range?

Reading it:

```
If point is at (3 mg/g actual, 2.9 mg/g predicted):
  → Error = 3 - 2.9 = +0.1 mg/g (slight underestimate)

If most points are ABOVE the line:
  → Model systematically underestimates

If most points are BELOW the line:
  → Model systematically overestimates
```

Panel 2: Residuals vs Predicted

```
y-axis: Residuals (mg/g)
x-axis: Predicted q (mg/g)

Good plot:
  • Random scatter around zero line
  • No patterns or trends
  • Constant variance (width is same everywhere)

Bad plot:
  • Funnel shape (variance increases with prediction)
  • Curved pattern (non-linear effects missed)
  • Systematic deviations (positive at low, negative at high)
```

What it tells you: - Do residuals depend on prediction magnitude? - Is there constant variance (homoscedasticity)? - Are there systematic patterns?

Reading it:

```
If you see a funnel:
  → Heteroscedasticity (not ideal but common)
  → Model is more certain at some ranges than others

If you see curved pattern:
  → Model may be missing non-linear effects
  → Might need higher-degree polynomial

If random scatter:
  → Model assumptions are satisfied
```

Panel 3: Residual Distribution

```
y-axis: Frequency (count)
x-axis: Residuals (mg/g)

Good plot:
  • Bell curve shape (normal distribution)
  • Centered at zero
  • Tails don't have unusual outliers
```

```
Bad plot:
  • Skewed left or right
  • Multiple peaks (bimodal)
  • Heavy tails (extreme outliers)
```

What it tells you: - Are residuals normally distributed? - Are there outliers? - Is there skewness?

Reading it:

```
If bell curve is centered at zero:
  → Model doesn't systematically over/underestimate

If skewed right (tail to right):
  → Model tends to underestimate some conditions

If outlier on left:
  → Some condition gives unexpectedly high residual
  → Worth investigating what's special about that point
```

Panel 4: Q-Q Plot

```
y-axis: Sample quantiles (residuals)
x-axis: Theoretical quantiles (normal distribution)

Good plot:
  • Points form roughly straight line
  • Minor deviations at tails (OK)

Bad plot:
  • S-shape (non-normal)
  • Curved (fat tails)
  • Staircase pattern (discrete values)
```

What it tells you: - Do residuals follow normal distribution? - Are there heavy tails? - How much do they deviate from normal?

Reading it:

```
Q-Q plot compares your residuals to what perfectly
normal distribution would look like.

If points follow line:
  → Residuals are approximately normal

If points deviate at tails:
  → May have outliers or heavy tails
  → Usually not a dealbreaker (normality is assumption, not requirement)
```

HOW TO INTERPRET RESULTS

The R^2 Value (Most Important)

Your expected value: R^2 0.84-0.87

What it means:

```
R² = 0.85 means:
  - Langmuir model explains 85% of the variation in q_removal
  - 15% is left unexplained (this is what ML will learn)
  - The remaining 15% comes from:
    * Non-linearities Langmuir doesn't capture
    * Interactions between factors
    * Mechanisms not included in model
```

How good is 0.85?

```
For a chemical equilibrium model:
  R² < 0.70 = Poor (model missing major mechanisms)
  R² = 0.75-0.80 = Good (captures main effects)
  R² = 0.85-0.90 = Very Good (our expectation) ← YOU ARE HERE
  R² > 0.95 = Exceptional (nearly perfect fit)

For ML residual predictions:
  R² < 0.85 = More work for ML
  R² = 0.90-0.93 = Good for ML to improve on
  R² > 0.95 = Limited room for ML improvement
```

Bottom line: R^2 0.85 is EXCELLENT for Phase 2! - Shows Langmuir is appropriate - Leaves enough residual variation for ML to learn - Provides good baseline for Phase 5 hybrid model

The RMSE Value (Practical Significance)

Your expected value: RMSE 0.9-1.2 mg/g

What it means:

```
RMSE = 0.92 mg/g means:
  - On average, predictions are off by ±0.92 mg/g
  - 68% of predictions are within ±0.92 mg/g
  - 95% are within ±1.84 mg/g
```

Is this good?

```
Your data range: 1.64 - 8.32 mg/g (span = 6.68)
Your RMSE: 0.92 mg/g
Percentage: 0.92/6.68 = 13.8% of range

Interpretation:
  RMSE < 10% of range = Good
  RMSE = 10-20% of range = Acceptable
  RMSE > 20% of range = Poor
```

Practical meaning:

```
Predictions: "Under these conditions, q  4.2 mg/g"
Actual: 3.1 mg/g
Error: 1.1 mg/g (within ±0.92 baseline noise)

For water treatment design:
  - If you want 4 mg/g removal, model says: "will be 3.1-5.3 mg/g"
  - That's a useful range, not a point estimate!
```

The Residual Distribution

What to look for:

1. **Mean 0:** (your model is unbiased)
2. **Normal distribution:** (good for statistics)
3. **Std dev 0.92:** (consistent with RMSE)
4. **No patterns:** (residuals are random)
5. **No outliers:** Acceptable (a few OK)

If you see:

```
Residual mean = -0.05:
  → Systematic slight underestimation
  → Could adjust intercept, but not critical

Residual std = 2.0 (instead of 0.9):
  → High variability
  → Check for outliers or non-linearities

Residuals follow curve in Residual Plot:
  → Missing non-linear effects
  → Might use higher polynomial degree
  → Or: this is fine for Phase 2, ML will learn it
```

DIAGNOSTIC CHECKS

Check 1: R^2 Adequacy

Question: Is R^2 in the expected range?

```
Expected: 0.80-0.90
Your value: 0.8456

  PASS - Right in the zone
```

If too low ($R^2 < 0.75$):

```
Possible reasons:
1. Data quality issue (check q_removal values)
2. Model too simple (but 66 features should be enough)
```

```
3. Strong non-linearity (degree-3 polynomial)
4. Missing important factors

Action:
  → Check for missing values or outliers
  → Plot residuals to find patterns
  → Try degree-3 polynomial (more complex)
```

If too high ($R^2 > 0.95$):

```
Possible reasons:
1. Overfitting (too many features)
2. Simulation quality (data is "too clean")
3. Real mechanisms well-captured

Action:
  → Do cross-validation test
  → Try degree-1 (just main effects) to compare
  → Don't worry - high R² at Phase 2 is OK!
```

Check 2: Residual Normality

Question: Are residuals normally distributed?

How to check:

```
Look at Panel 4 (Q-Q plot):
  - Points form straight line? → Normal
  - Points deviate at tails? → Slight non-normality (OK)
  - S-shaped pattern? → Significant non-normality (concern)

Look at Panel 3 (Residual Distribution):
  - Bell curve? → Normal
  - Skewed? → Slight issue
  - Multiple peaks? → Real problem
```

If residuals are non-normal:

```
Severity: Low (linear regression is robust to non-normality)

Action:
  → Check for outliers (|residual| > 2 × std)
  → Check for skewness (positive or negative tail)
  → Log-transform response? (try: y_new = log(y))
  → Use robust regression? (weights outliers less)
```

Check 3: Heteroscedasticity (Constant Variance)

Question: Is variance of residuals constant?

How to check:

```
Look at Panel 2 (Residuals vs Predicted):
  - Constant width? → Homoscedastic
  - Funnel shape? → Heteroscedastic
  - Growing/shrinking? → Variance changes
```

If you see funnel (heteroscedasticity):

```
Meaning:
  - Model is more certain at low values
  - Less certain at high values
  - Or vice versa

Severity: Medium (violates assumption, but common)

Action:
  → Weighted least squares (weight by 1/variance)
  → Log-transform response (often helps)
  → Use robust regression
  → Or: accept it - Phase 2 is baseline anyway
```

Check 4: Outliers

Question: Are there extreme residuals?

How to check:

```
Rule: If |residual| > 2.5 × std, it's an outlier

Your std = 0.92:
  2.5 × 0.92 = 2.3 mg/g

Look at results CSV:
  - Residuals outside ±2.3 mg/g = Outliers
  - Check what's special about those conditions
```

If you find outliers:

```
Action:
1. Investigate the condition:
   - What pH, concentration, ions, etc.?
   - Is it an extreme combination?
   - Is it physically reasonable?

2. Decide:
   - Keep it: Maybe real non-linearity
   - Remove it: If measurement error suspected
   - Note it: For Phase 3 analysis

3. Document:
   - Which point (Run number)?
   - What was the error?
```

- Why might it occur?

TROUBLESHOOTING

Issue 1: R^2 Too Low (< 0.75)

Symptom:

```
R² = 0.65 (not 0.85)
Model Performance Metrics:
R² (Coefficient of Determination): 0.6512 (65.12%)
```

Possible causes:

Cause	How to check	How to fix
Data quality	Check q_removal for NaN/inf/outliers	Clean data, remove bad points
Wrong features	Plot q_removal vs each factor	Verify relationships exist
Need non-linearity	Look at residual plots	Use degree-3 polynomial
Scale issue	Check StandardScaler	Ensure both X and y scaled

Diagnosis steps:

```python
# Step 1: Check for bad data
print(results_df['q_removal'].describe())
print(results_df['q_removal'].isna().sum())  # Any NaN?
print(results_df['q_removal'].isin([inf, -inf]).sum())  # Any inf?

# Step 2: Check feature relationships
import matplotlib.pyplot as plt
for col in ['pH', 'C0', 'Time', 'Dose']:
    plt.scatter(results_df[col], results_df['q_removal'])
    plt.xlabel(col)
    plt.ylabel('q_removal')
    plt.show()

# Step 3: Try higher-order polynomial
poly = PolynomialFeatures(degree=3)  # Instead of degree=2
```

Issue 2: R^2 Too High (> 0.95)

Symptom:

```
R² = 0.98 (suspiciously good)
Model seems too good to be true
```

Possible causes:

Cause	Likelihood	Fix
Overfitting	Low (500 samples, 66 features is OK)	Check CV score
Simulation quality	High (synthetic data is cleaner than real)	Expected - OK for Phase 2
Data leakage	Low (checked data structure)	Verify no cheating occurred

Is this a problem?

```
No, actually good for Phase 2!
- Shows Langmuir captures the physics well
- Leaves room for ML to learn residuals (if they're random)
- Only concern: if residuals have patterns (model missing something)

Action: Check residual plots
  - If random scatter: Perfect! All good.
  - If patterns: Model missing something, investigate.
```

Issue 3: Residuals Not Normal

Symptom:

```
Shapiro-Wilk test p-value = 0.002 (< 0.05, reject normality)
Q-Q plot shows S-shape
```

Severity: Low (regression is robust)

Possible causes:

Cause	Sign	Fix
Outliers	Large residuals at tails	Remove extreme points
Skewness	One long tail	Log-transform response
Non-linearity	Curved pattern	Higher-degree polynomial

Action:

```python
# Check for outliers
outlier_threshold = 2.5 * residuals.std()
outliers = abs(residuals) > outlier_threshold
print(f"Outliers: {outliers.sum()} points")

# Try log-transform
y_log = np.log(y)
# Refit and check normality

# Or accept it
# Linear regression is robust to non-normality
# Don't worry too much for Phase 2
```

Issue 4: Heteroscedasticity (Funnel)

Symptom:

```
Residuals vs Predicted plot shows funnel shape
Errors are large at high predictions, small at low
```

Severity: Medium

Why it happens:

```
Model coefficients are same for all values
But relative error is proportional to magnitude
Example: Error of ±0.9 mg/g is bigger % at 3 mg/g than 8 mg/g
```

Fix options:

```python
# Option 1: Weighted least squares
from sklearn.linear_model import Ridge
linreg = Ridge(alpha=0.1)  # Slight regularization
linreg.fit(X_poly, y_scaled)

# Option 2: Log-transform response
y_log = np.log(y)
# Fit on log scale, inverse-transform for results
y_pred_log = linreg.predict(X_poly)
y_pred = np.exp(y_pred_log)

# Option 3: Accept it
# It's not ideal but acceptable for Phase 2 baseline
# ML will learn to handle it in Phase 4
```

Issue 5: Script Crashes

Error: `ImportError: No module named 'sklearn'`

Fix:

```
pip install scikit-learn scipy matplotlib
```

Error: `FileNotFoundError: data/dataset_simulated_500.csv`

Fix:

```
# Make sure file exists
ls data/dataset_simulated_500.csv

# If not, copy it
cp /path/to/file data/dataset_simulated_500.csv
```

Error: `MemoryError`

Fix:

```
# Dataset is only 500 rows - shouldn't happen
# But if it does, check RAM usage:
import psutil
print(psutil.virtual_memory())

# Or: Reduce polynomial degree
poly = PolynomialFeatures(degree=1)  # Linear only
```

WHAT COMES NEXT (PHASE 3)

Preview: Residual Analysis

Phase 2 leaves you with: - Langmuir baseline ($R^2 = 0.85$) - Residuals (what chemistry missed) - Model fit (interpretable)

Phase 3 will: 1. Analyze residual patterns 2. Identify what Langmuir missed 3. Engineer features to capture residuals 4. Prepare for ML training

Preview: ML Training (Phase 4)

Phase 4 will: 1. Train Random Forest on residuals 2. Train XGBoost on residuals 3. Train MLP Neural Network 4. Use cross-validation (5-fold) 5. Select best model

Expected result: - R^2 0.90-0.93 on residuals - Hybrid R^2 0.94-0.96 when combined with Langmuir

Preview: Hybrid Integration (Phase 5)

Phase 5 will: 1. Take Langmuir predictions: q_lang 2. Take ML predictions on residuals: Δq_ml 3. Combine: q_hybrid = q_lang + Δq_ml 4. Validate: Check R^2 0.94 achieved!

SUMMARY

What Langmuir Fitting Does

```
Takes:  500 data points with factors and responses
Does:   Fit multi-factor Langmuir model with polynomial features
Returns: R²  0.85, residuals, diagnostic plots
Means:  Chemical model captures 85% of fluoride removal patterns
```

The Model

```
q = (Linear regression on 66 polynomial features)
  = (  +  ×pH +  ×C0 + ... +  ×interaction terms)

Implicitly learns:
  - How each factor affects qmax and KL
  - 2-way interactions (time×pH, dose×concentration, etc.)
  - Non-linear effects (pH bell curve, concentration saturation)
```

Expected Results

```
R² = 0.84-0.87  (Excellent for chemistry model)
RMSE = 0.9-1.2 mg/g (13-18% of data range)
Residuals = Normal, random, centered at 0
Interpretation = Langmuir appropriate, ML can improve 15%
```

Why This Matters

```
Phase 2 establishes:
  1. Baseline performance (R² = 0.85)
  2. What residuals look like (opportunities for ML)
  3. That factors matter (polynomial features improve fit)
  4. Validation of chemical model (Langmuir is right choice)

It sets up Phase 3-5:
  → Phase 3: Understand residual patterns
  → Phase 4: ML learns to predict residuals
  → Phase 5: Hybrid achieves R²  0.94
```

CONCLUSION

Langmuir fitting is the **bridge between chemistry and machine learning**:

- Chemistry tells us the form of the equation
- Data tells us the parameters
- Residuals tell us where chemistry fails
- ML learns to fix those failures

Your Phase 2 goal: Fit the Langmuir model and understand the residuals.

Your Phase 5 goal: Combine Langmuir + ML for R^2 0.94.

Ready? Run the script and begin Phase 2!

End of Comprehensive Guide

Total Length: 15,000+ words **Covers:** Theory, implementation, interpretation, troubleshooting, and next steps **For:** Deep understanding of Phase 2 Langmuir fitting

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